

Fundamental Calculus/Differential and Integral Calculus

Prepared By : Nasrina Parvin
Lecturer, INS, UIU

Reference : Calculus 10-th Edition by Howard Anton,
Irl Bivens and Stephen Davis

Chapter: 0.4

INVERSE

FUNCTIONS(Page-38)

INVERSE FUNCTIONS(Page-40)

- *In mathematics the term inverse is used to describe functions that reverse one another in the sense that each undoes the effect of the other. In an inverse function Independent variable becomes dependent variable and dependent variable becomes Independent variable*
- *Domain of function $f(x)$ = Range of function $f^{-1}(x)$ and*
- *Range of function $f(x)$ = Domain of function $f^{-1}(x)$*

0.4.2 theorem *If an equation $y = f(x)$ can be solved for x as a function of y , say $x = g(y)$, then f has an inverse and that inverse is $g(y) = f^{-1}(y) \Rightarrow g(x) = f^{-1}(x)$*

The Horizontal Line Test

0.4.3 THEOREM *A function has an inverse if and only if it is one-to-one.*

0.4.4 theorem (*The Horizontal Line Test*)

A function has an inverse function if and only if its graph is cut at most once by any horizontal line.

Many to one function , $y = x^2$

Input x	Output $y = x^2$
$x = -\infty$	$y = +\infty$
$x = -2$	$y = 4$
$x = -1$	$y = 1$
$x = 1$	
$x = 2$	
$x = +\infty$	

The diagram illustrates the many-to-one function $y = x^2$. The table shows the mapping from input x to output y . Arrows indicate the mapping from x to y , showing that different inputs can result in the same output (e.g., $x = -1$ and $x = 1$ both map to $y = 1$).

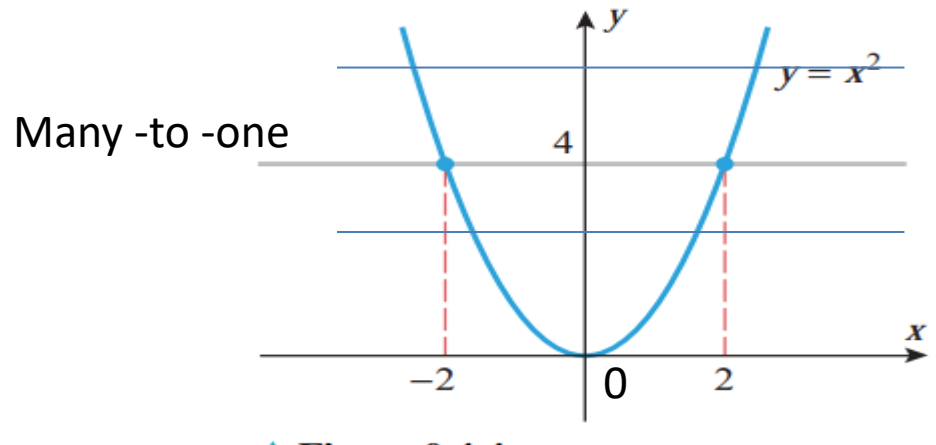
one to one function $y = x^3$

Input x	Output $y = x^3$
$x = -\infty$	$y = -\infty$
$x = -2$	$y = -8$
$x = -1$	$y = -1$
$x = 0$	$y = 0$
$x = 1$	$y = 1$
$x = 2$	$y = 8$
$x = +\infty$	$y = +\infty$

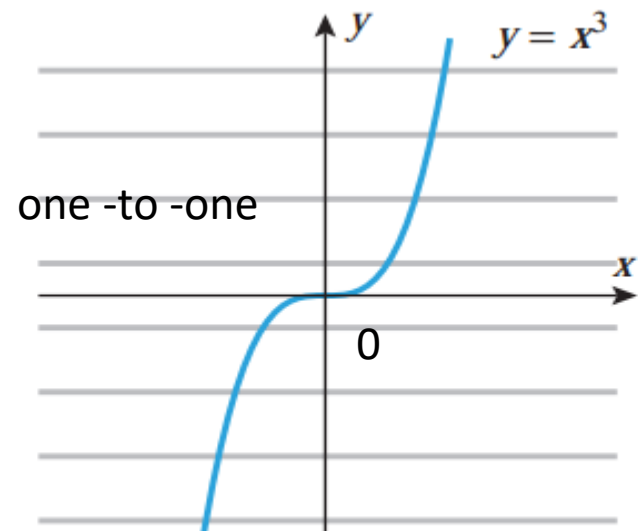
The Horizontal Line Test

- Determine whether f is one-to-one or many to-one (i) $y = x^2$ (ii) $y = x^3$

(i) Since from horizontal line test we can say that, for $y = x^2$ the horizontal line cuts or intersects the graph more than one point, so the given function is Many-to-one. So function has no inverse function.



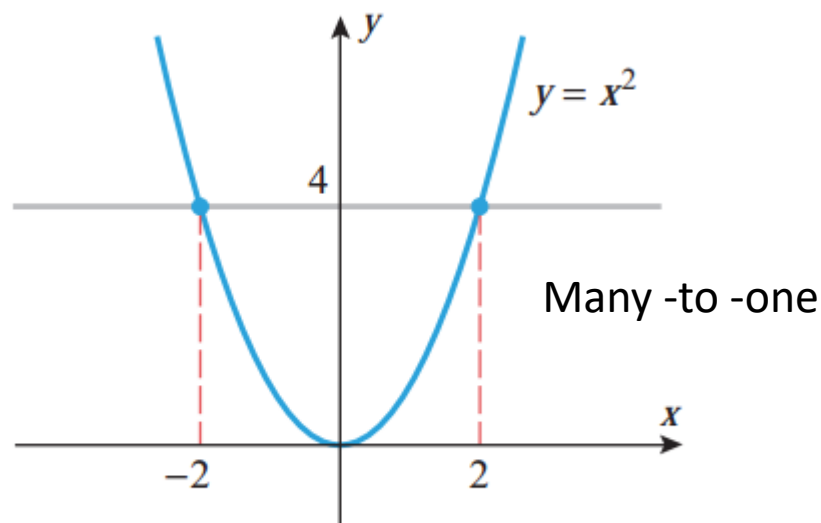
(ii) Since from horizontal line test we can say that, for $y = x^3$ the horizontal line cuts or intersects the graph only at one point, so the given function is One-to-one. So function has inverse function.



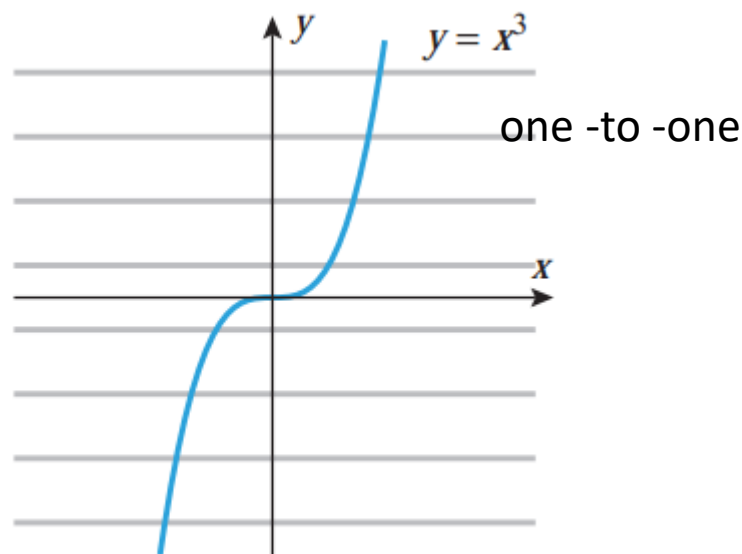
The Horizontal Line Test

► **Example 5** Use the horizontal line test to show that $f(x) = x^2$ has no inverse but that $f(x) = x^3$ does.

Solution. Figure 0.4.4 shows a horizontal line that cuts the graph of $y = x^2$ more than once, so $f(x) = x^2$ is not invertible. Figure 0.4.5 shows that the graph of $y = x^3$ is cut at most once by any horizontal line, so $f(x) = x^3$ is invertible. [Recall from Example 2 that the inverse of $f(x) = x^3$ is $f^{-1}(x) = x^{1/3}$.] ◀



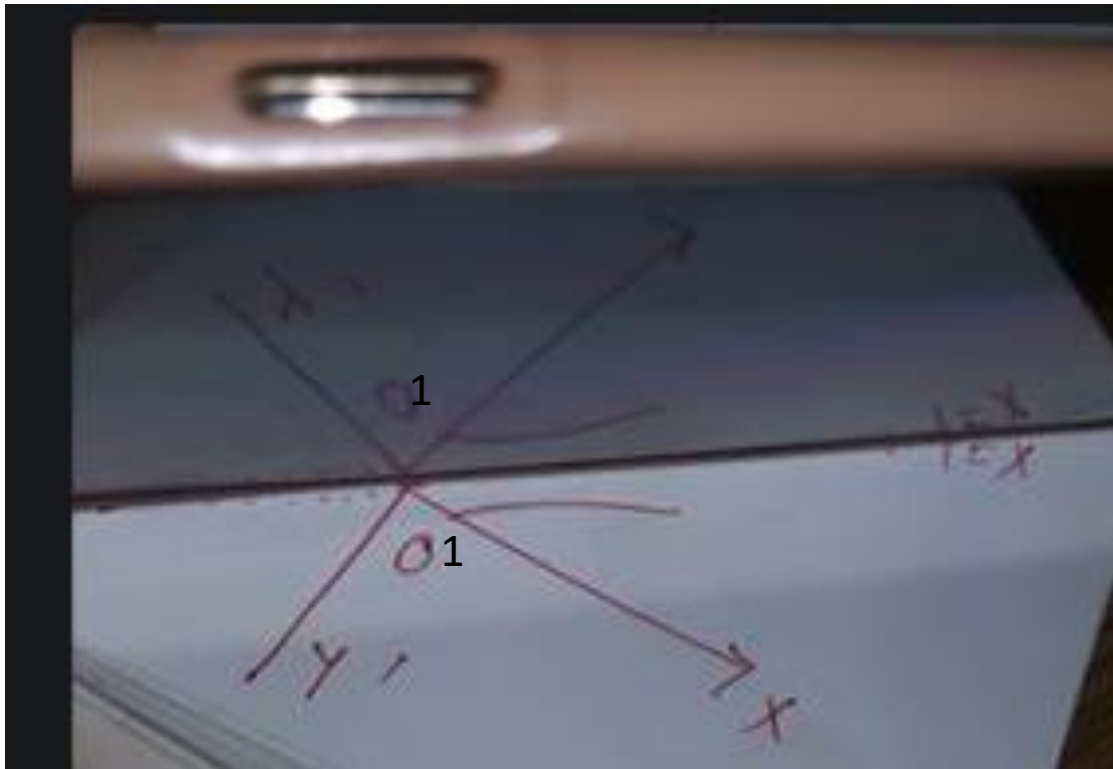
▲ Figure 0.4.4



▲ Figure 0.4.5

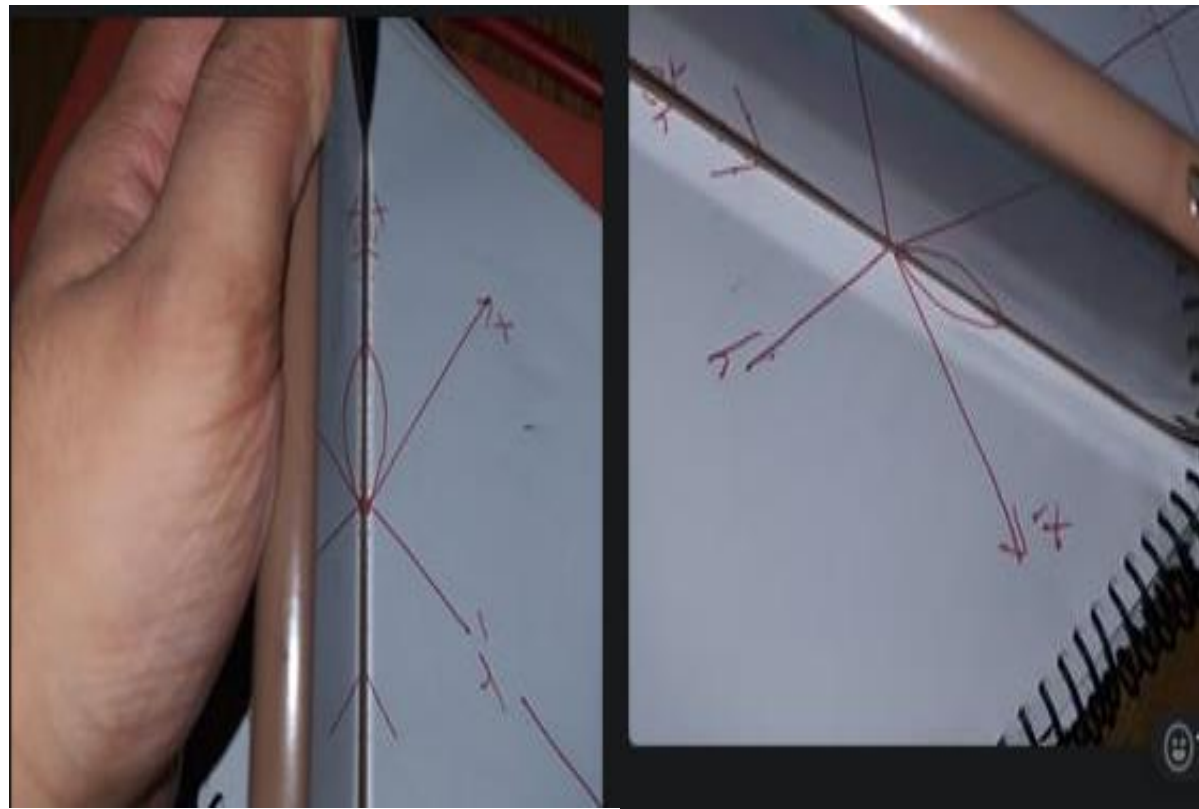
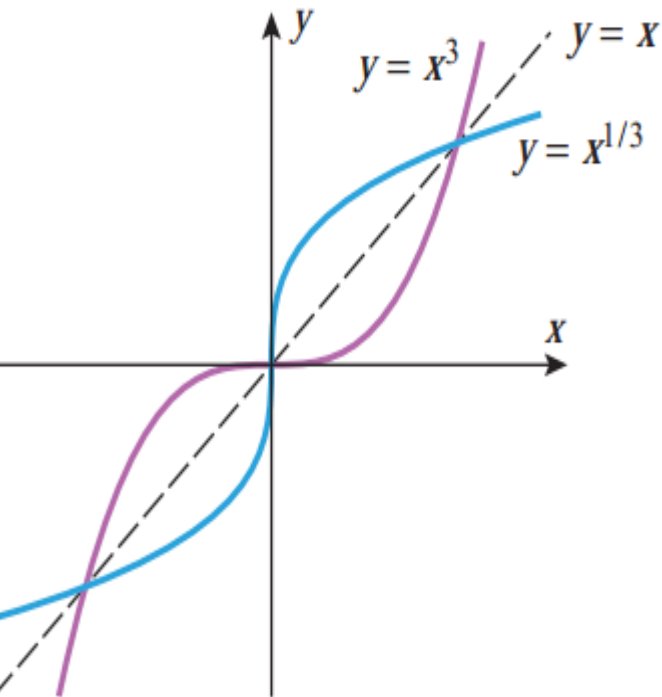
INVERSE FUNCTIONS

0.4.5 THEOREM *If f has an inverse, then the graphs of $y = f(x)$ and $y = f^{-1}(x)$ are reflections of one another about the line $y = x$; that is, each graph is the mirror image of the other with respect to that line.*



INVERSE FUNCTIONS

0.4.5 THEOREM *If f has an inverse, then the graphs of $y = f(x)$ and $y = f^{-1}(x)$ are reflections of one another about the line $y = x$; that is, each graph is the mirror image of the other with respect to that line.*



A Procedure for Finding the Inverse of a Function f

- Step 1.

- Write down the equation $y = f(x)$.

- Step 2.

- If possible, solve this equation for x as a function of y .

- Step 3.

- The resulting equation will be $x = f^{-1}(y)$, which provides a formula for f^{-1} with y as the independent variable.

- Step 4.

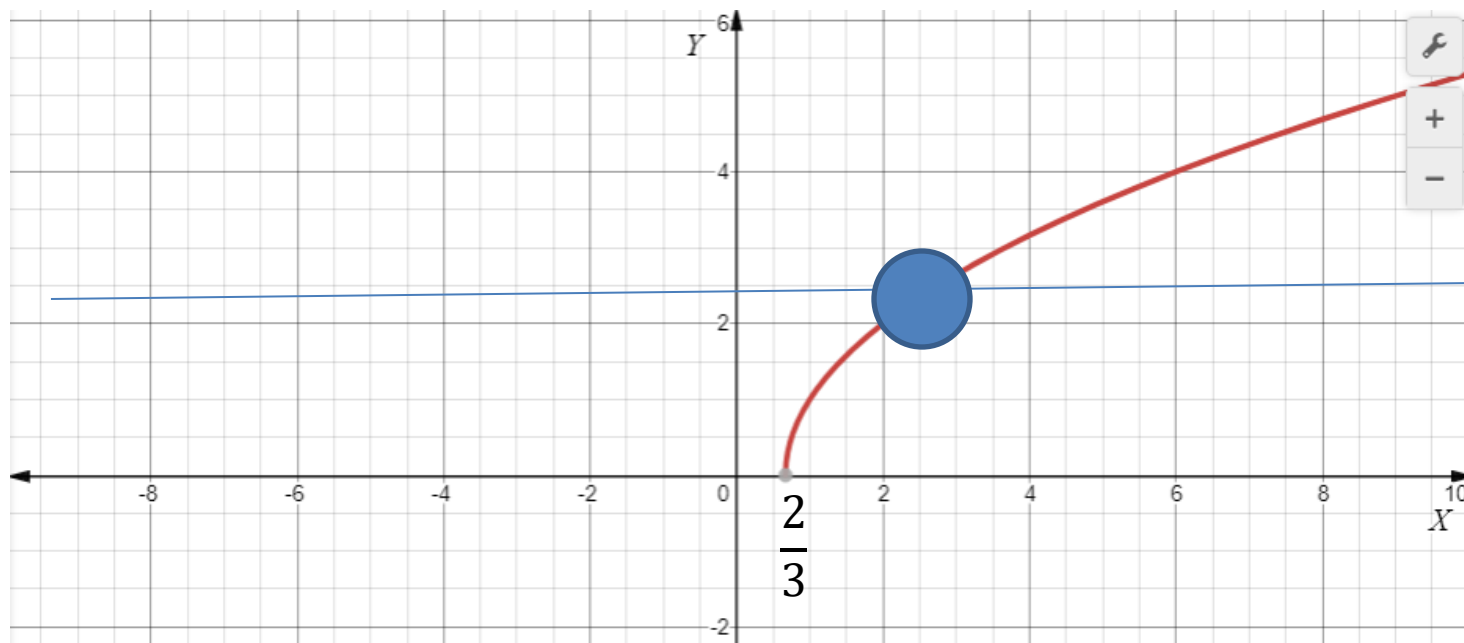
- If y is acceptable as the independent variable for the inverse function, then you are done, but if you want to have x as the independent variable, then you need to interchange x and y in the equation $x = f^{-1}(y)$ to obtain $y = f^{-1}(x)$.

Finding the Inverse of a Function f

- Determine whether f is one-to-one or many to-one, find its inverse functions (if -possible). draw the graph of $f(x)$ and its inverse $f^{-1}(x)$ in the same diagram, where
- (i) $f(x) = \sqrt{3x - 2}$ (ii) $f(x) = x^3 + 1$, (iii) $f(x) = \sqrt[3]{x - 1}$
also find their domain and range.

Finding the Inverse of a Function f

- (i) Given, $f(x) = \sqrt{3x - 2}$
- Since from horizontal line test we can say that, the horizontal line cuts or intersects the graph only at one point, so the given function is one-to-one function.



Finding the Inverse of a Function f

So , $f(x)$ has an inverse function.

$$\text{Given, } f(x) = y = \sqrt{3x - 2}$$

$$\Rightarrow y = \sqrt{3x - 2}; 3x - 2 \geq 0$$

$$\Rightarrow 3x \geq 2 \Rightarrow x \geq \frac{2}{3};$$

Calculation: Finding inverse function

$$y = \sqrt{3x - 2} \Rightarrow y^2 = 3x - 2$$

$$\Rightarrow 3x = y^2 + 2$$

$$\Rightarrow x = \frac{y^2 + 2}{3}$$

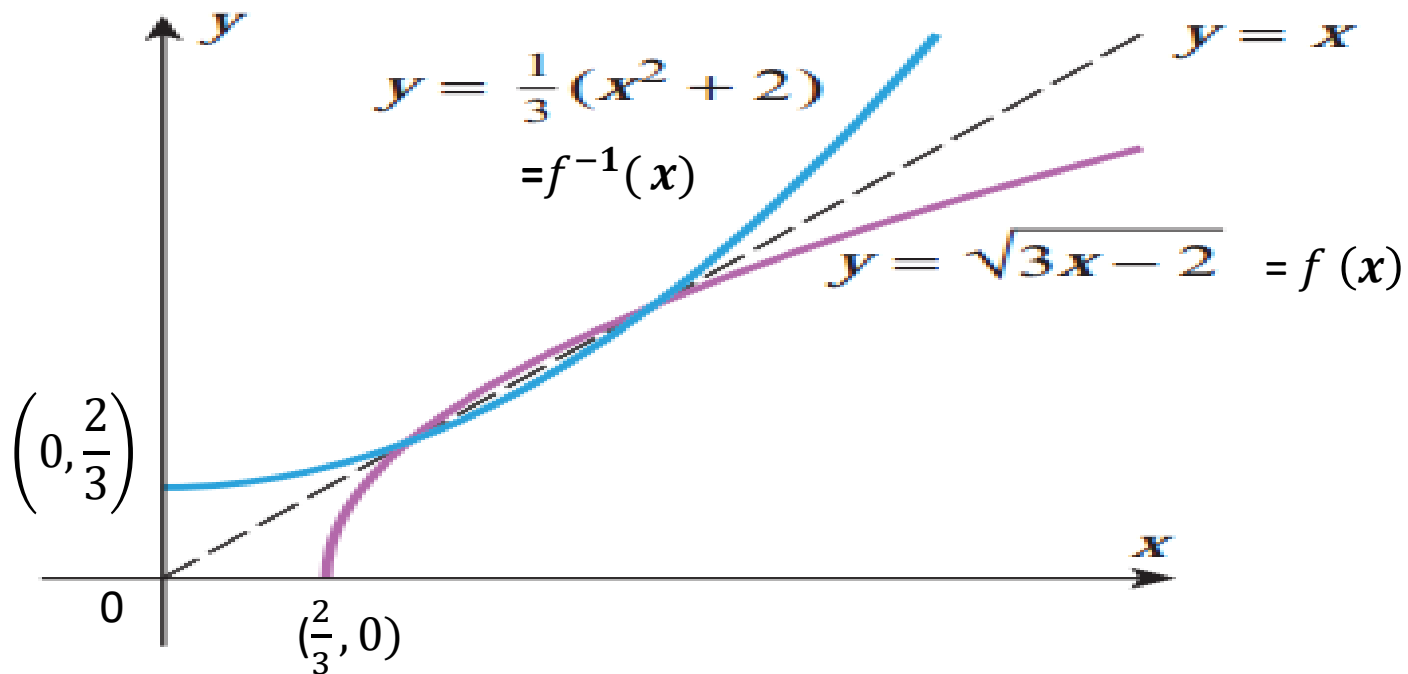
$$\Rightarrow x = f^{-1}(y) = g(y) = \frac{y^2 + 2}{3} \Rightarrow f^{-1}(x) = g(x) = \frac{x^2 + 2}{3}$$

INVERSE FUNCTIONS

Graph of $f(x)$ and its inverse $f^{-1}(x)$ in the same diagram,

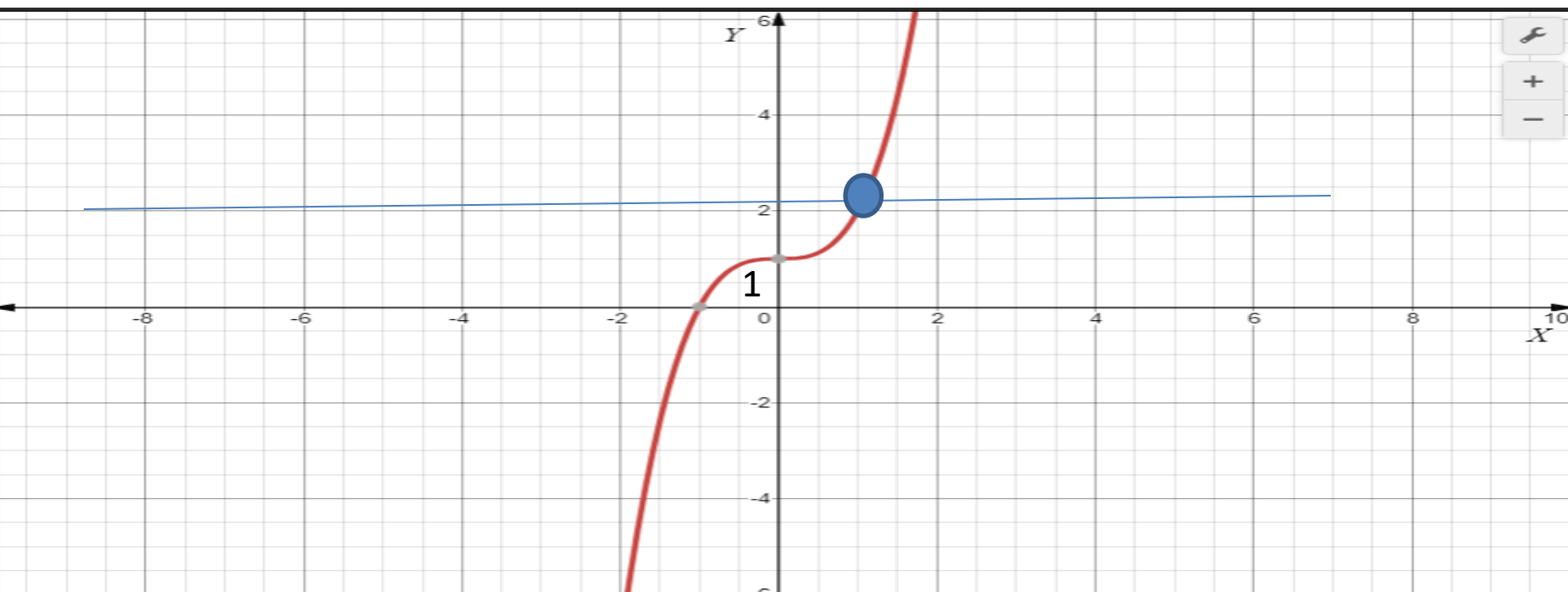
Domain of $f(x)$ is $[\frac{2}{3}, +\infty)$, Range of f is $[0, +\infty)$ and

Domain of $f^{-1}(x)$ is $[0, +\infty]$, Range of $f^{-1}(x)$ is $[\frac{2}{3}, +\infty)$.



Finding the Inverse of a Function f

- (i) Given, $f(x) = x^3 + 1$
- Since from horizontal line test we can say that, the horizontal line cuts or intersects the graph only one point, so the given function is one-to-one function.



Finding the Inverse of a Function f

So , $f(x)$ has an inverse function.

Calculation:

$$\text{Given, } f(x) = y = x^3 + 1$$

$$\Rightarrow y - 1 = x^3;$$

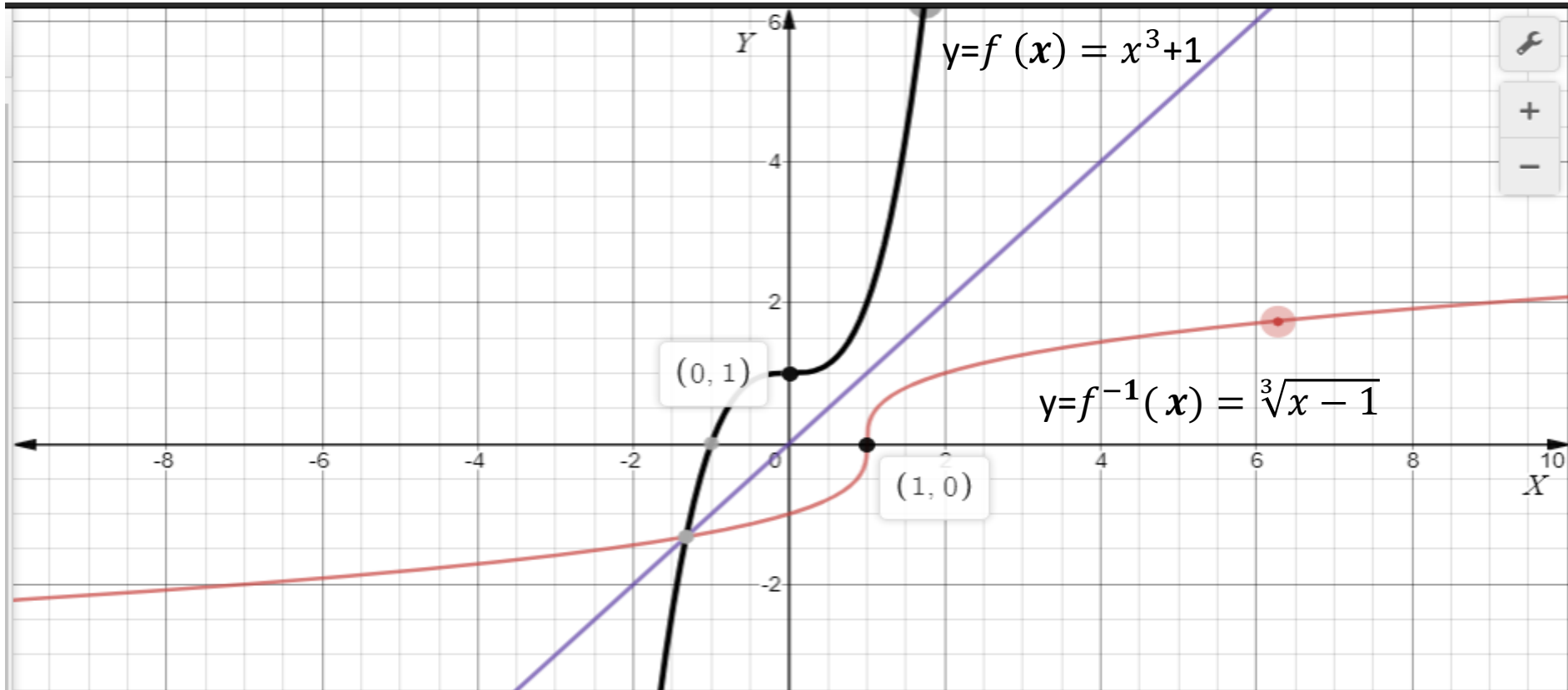
$$\Rightarrow x = \sqrt[3]{y - 1} \Rightarrow x = f^{-1}(y) = g(y) = \sqrt[3]{y - 1}$$

$$\Rightarrow f^{-1}(x) = g(x) = \sqrt[3]{x - 1}$$

INVERSE FUNCTIONS

Graph of $y = f(x) = x^3 + 1$

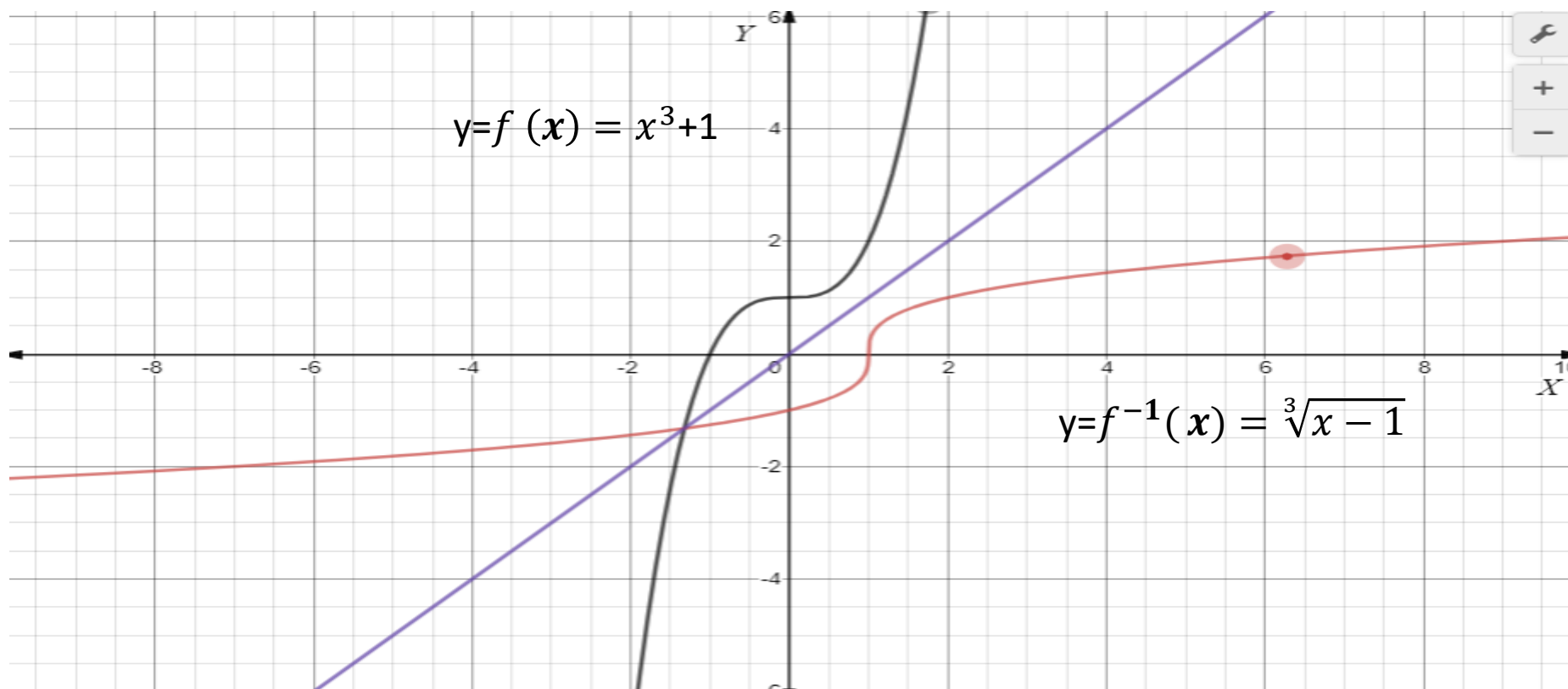
its inverse $y = f^{-1}(x) = \sqrt[3]{x - 1}$ in the same diagram,



INVERSE FUNCTIONS

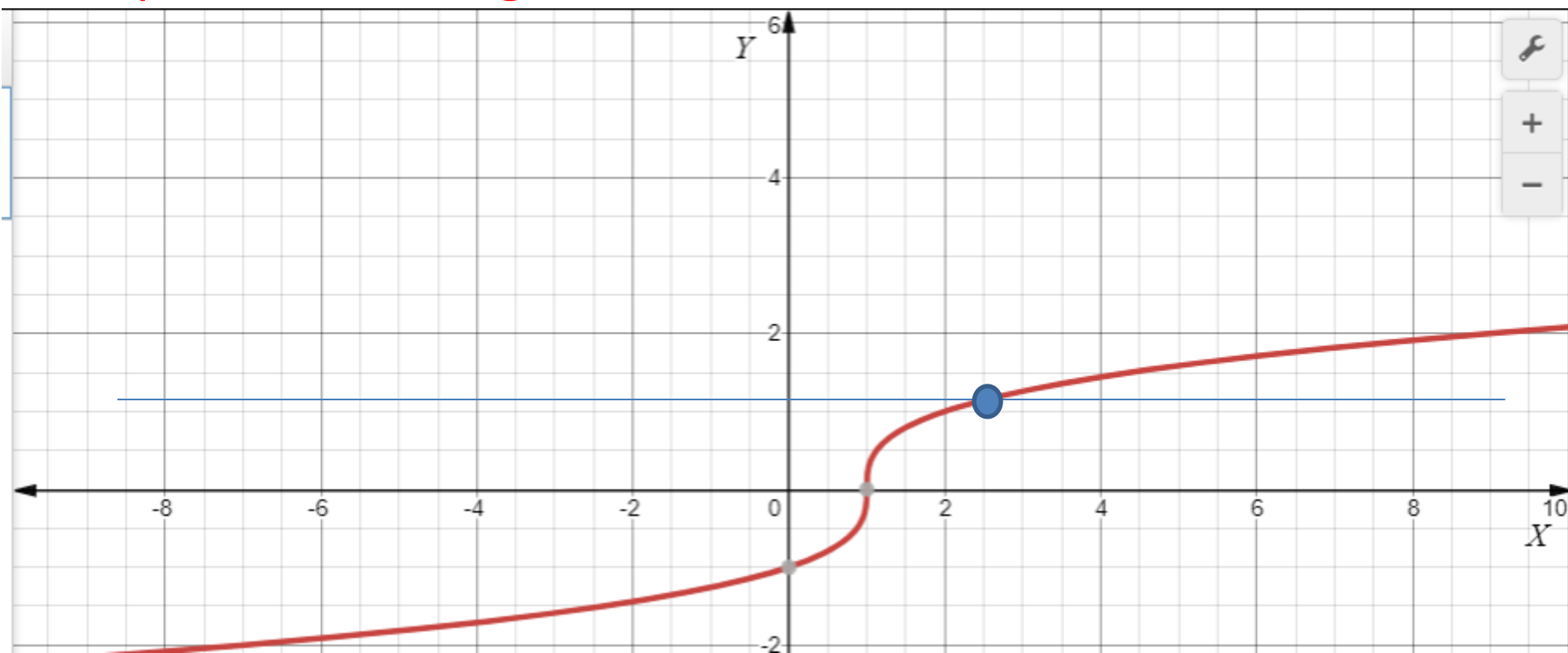
Graph of $y = f(x) = x^3 + 1$

its inverse $y = f^{-1}(x) = \sqrt[3]{x - 1}$ in the same diagram,



Finding the Inverse of a Function f

- (i) Given, $f(x) = \sqrt[3]{x-1}$ (Home work)
- Since from horizontal line test we can say that, the horizontal line cuts or intersects the graph only one point, so the given function is one-to-one function.



Finding the Inverse of a Function f

So , $f(x)$ has an inverse function.

Calculation:

$$\text{Given, } f(x) = y = \sqrt[3]{x-1}$$

$$\Rightarrow x - 1 = y^3;$$

$$\Rightarrow y^3 = x - 1$$

$$\Rightarrow x = y^3 + 1 \Rightarrow f^{-1}(y) = g(y) = y^3 + 1$$

$$\Rightarrow f^{-1}(x) = g(x) = x^3 + 1$$

INVERSE FUNCTIONS

Graph of $f(x) = \sqrt[3]{x-1}$
its inverse $f^{-1}(x) = x^3+1$ in the same diagram,

